ReImagine: Rethinking Controllable High-Quality Human Video Generation via Image-First Synthesis

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Research Questions

- How can we generate high-quality human videos with simultaneous control over body pose, appearance, and camera viewpoint without requiring large-scale multi-view video data?
- Can decoupling appearance modeling from temporal consistency via an image-first approach achieve better results than end-to-end video generation?
- How effective is training-free temporal refinement using pretrained video diffusion models for achieving temporal consistency?

Core Idea

The paper addresses the challenge that existing human video generation methods struggle with joint pose and viewpoint control due to limited multi-view training data. The key insight is to decouple the problem: first generate high-quality pose- and viewpoint-controlled images using a pretrained image backbone (FLUX) fine-tuned with LoRA, then achieve temporal consistency through training-free refinement with a pretrained video diffusion model. This avoids the need for large-scale multi-view video datasets while leveraging strong image generation priors.

Key Formula

Spatiotemporal Regularization

$$\tilde{\mathcal{F}}(\mathbf{h}_s) = \mathcal{F}(\mathbf{h}_s) \odot \exp\left(-\left(\frac{f_t}{\tau_t}\right)^2 - \left(\frac{f_x}{\tau_s}\right)^2 - \left(\frac{f_y}{\tau_s}\right)^2\right)$$

This applies 3D Fourier filtering to latent representations h_s during temporal refinement. f_t, f_x, f_y are temporal and spatial frequencies, τ_t=0.06 and τ_s=0.12 control smoothing strength. This suppresses high-frequency temporal artifacts while preserving spatial details, enabling training-free temporal consistency.

Condition-Aware Positional Encoding

$$\mathbf{p}_i = (x_i, y_i, c_i)$$

Each token i gets spatial coordinates (x_i, y_i) and condition index c_i (pose/front-view/back-view/noise). This allows the DiT to distinguish heterogeneous token types while preserving spatial structure. Applied via RoPE to enable joint processing of SMPL-X parameters, appearance tokens, and noise tokens.

Key Figure

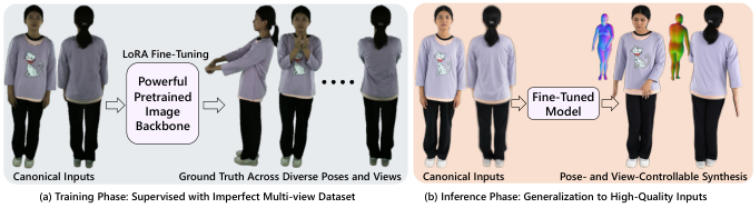

Fig. 2: Overview of our image-first training and inference paradigm. (a) During training, a powerful pretrained image backbone is fine-tuned via lightweight LoRA adaptation using an imperfect multi-view dataset with structured pose and viewpoint supervision. (b) At inference, the fine-tuned model generalizes to high-quality canonical inputs and enables pose- and viewpoint-controllable human synthesis. This highlights our key insight that strong pretrained image priors substantially reduce the need for large-scale, high-fidelity multi-view video data.

- We propose a pose- and viewpoint-controllable human video generation pipeline from canonical front–back images and SMPL-X motion sequences, achieving strong performance in temporally coherent and viewpoint-aware synthesis.
- We demonstrate that decoupling appearance modeling from temporal consistency, by combining a pretrained image backbone with training-free refinement using a pretrained video diffusion model, enables high-quality generation without task-specific video training under limited multi-view data.
- We construct a canonical human dataset with disentangled face, clothing, and footwear attributes, together with an end-to-end model for attribute-consistent full-body human image generation, both publicly released to support future research.

2 Related Works

2.1 Image-based Human Generation

The evolution of image-based human generation is characterized by a significant transition from adversarial learning [10, 15, 19–21] to latent diffusion paradigms [35, 36, 41]. Early research was dominated by Generative Adversarial Networks. Pioneering StyleGAN-Human [9] focus on unconditional human image generation, achieving photorealism through a minimax game between a generator and a discriminator. HumanGAN [38] is capable of performing all tasks related to global appearance sampling, pose transfer, part and garment transfer, as well as part sampling. Nevertheless, GANs often faced challenges with training instability and mode collapse, leading to the rise of Denoising Diffusion Probabilistic Models [27, 37, 41].

Diffusion models utilize an iterative denoising process to recover clean images from Gaussian noise, offering superior stability and diversity. And beyond

Figure 2: Shows the core insight that pretrained image backbones can be effectively fine-tuned on imperfect multi-view data to achieve high-quality pose and viewpoint control, avoiding the need for large-scale video training.

Limitations

- Requires canonical front-back reference images which may not always be available in practice
- Frame-wise generation introduces subtle appearance variations that require additional temporal refinement
- Performance degrades significantly when back-view appearance input is missing, as shown in ablation studies
- Limited evaluation on extremely diverse clothing styles and complex motion sequences